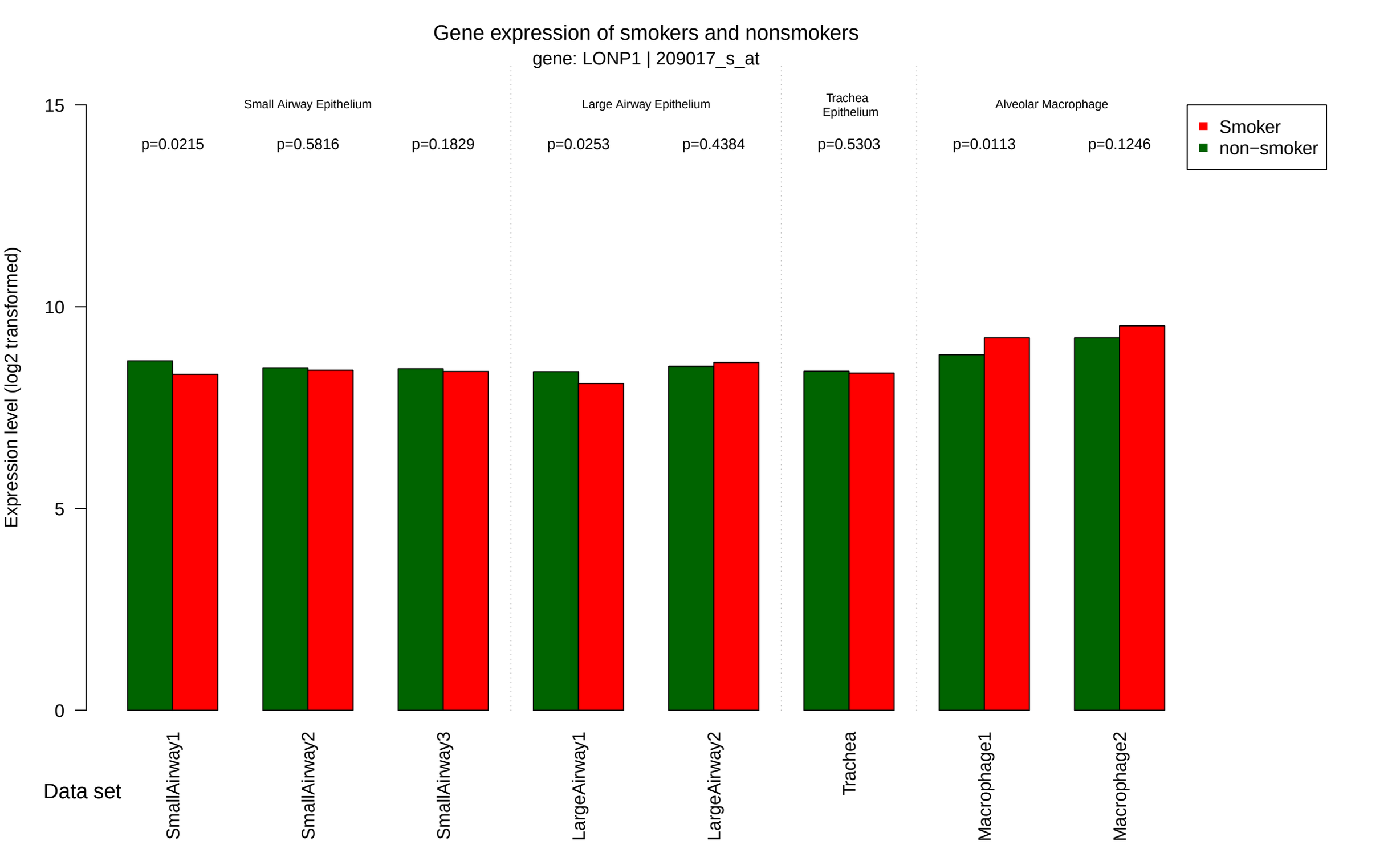
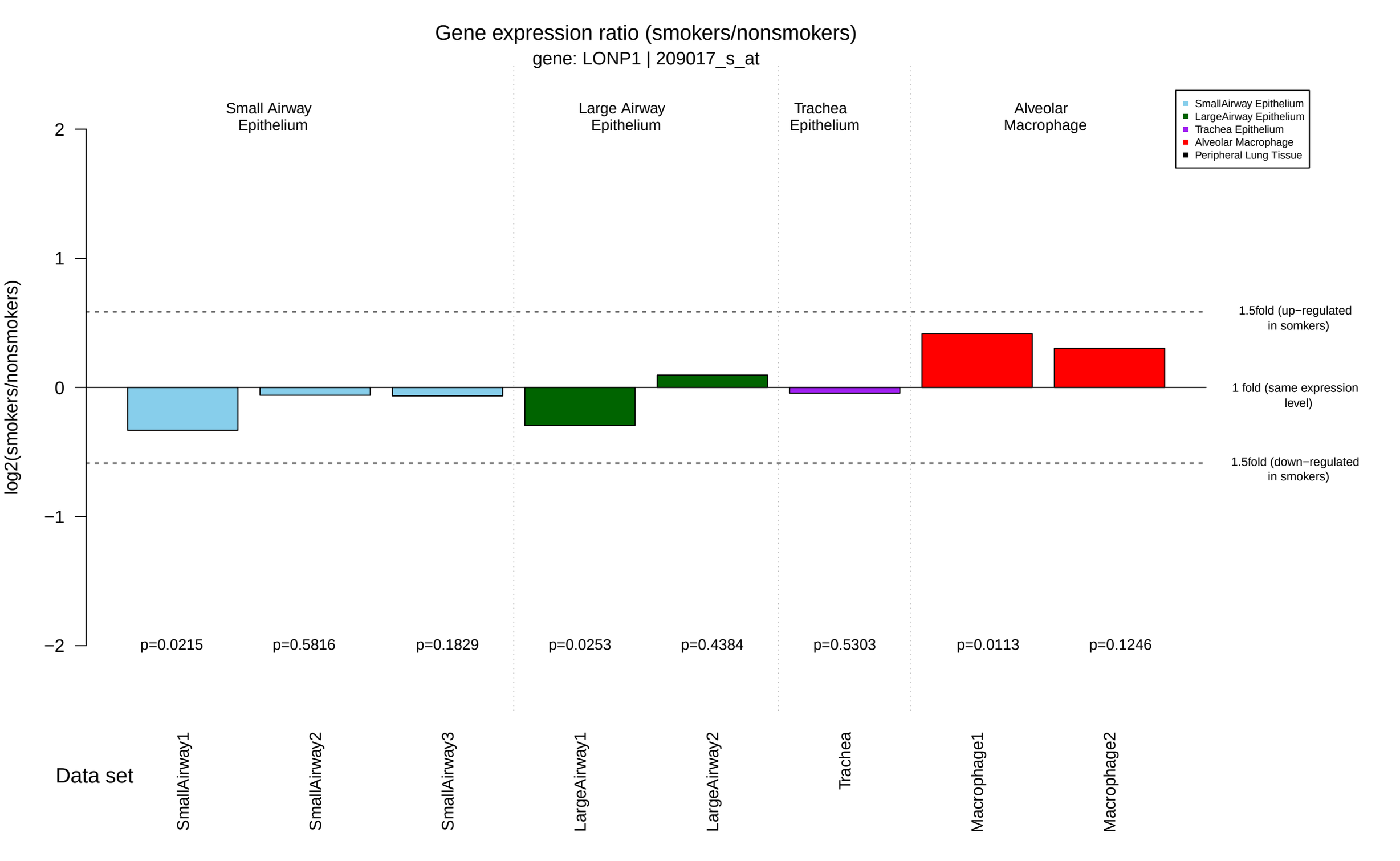
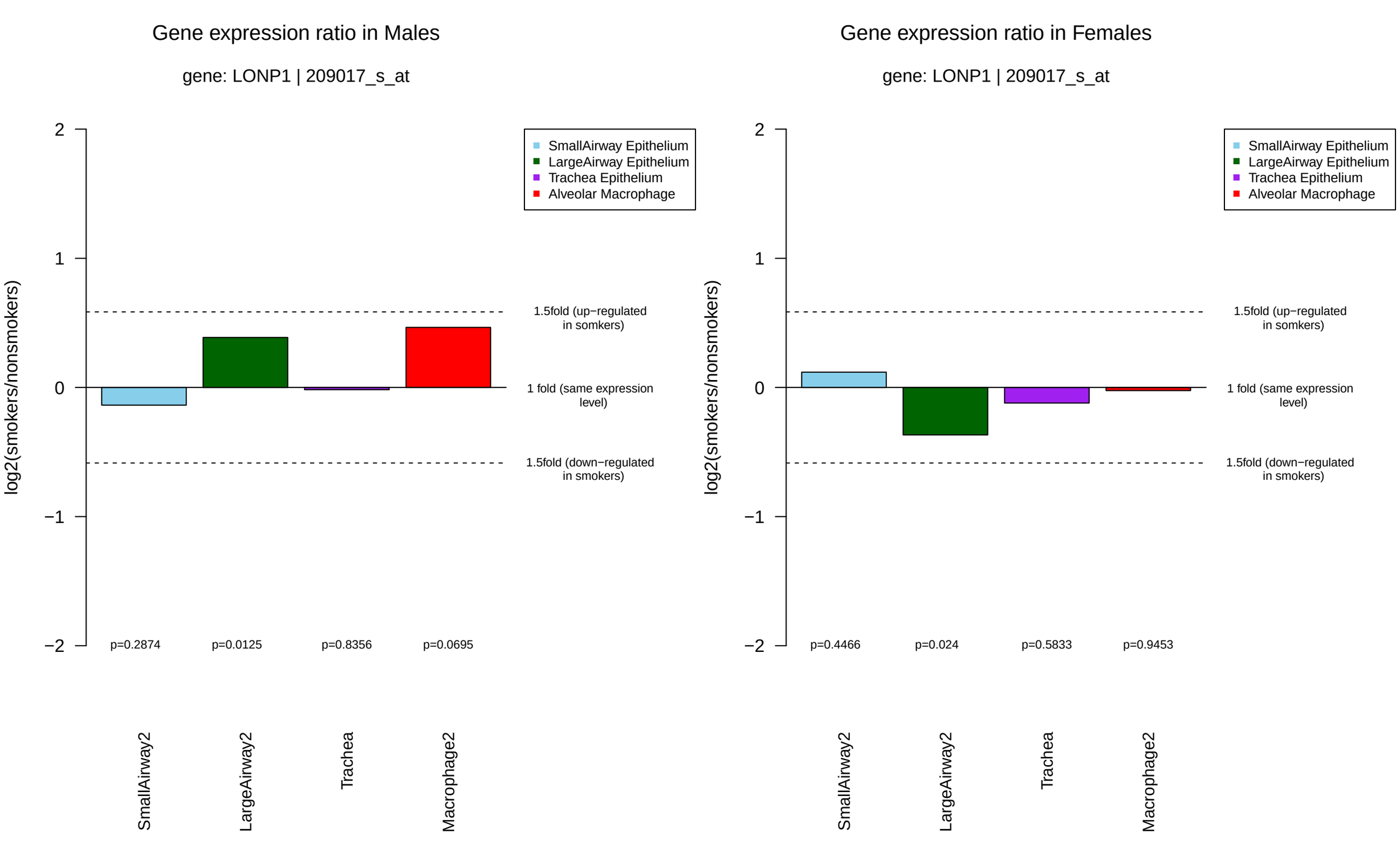
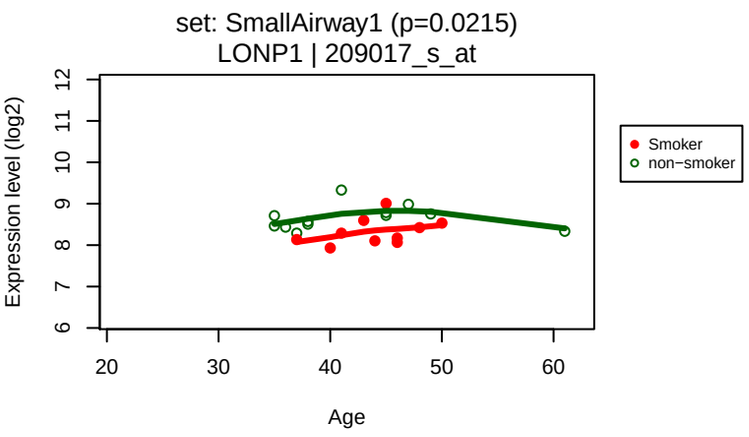
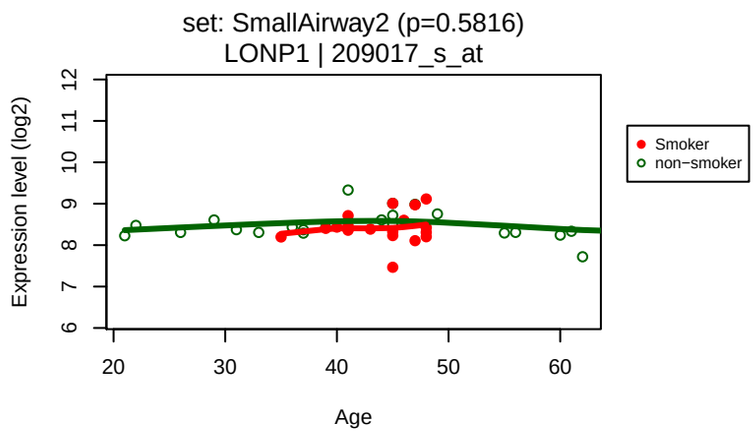
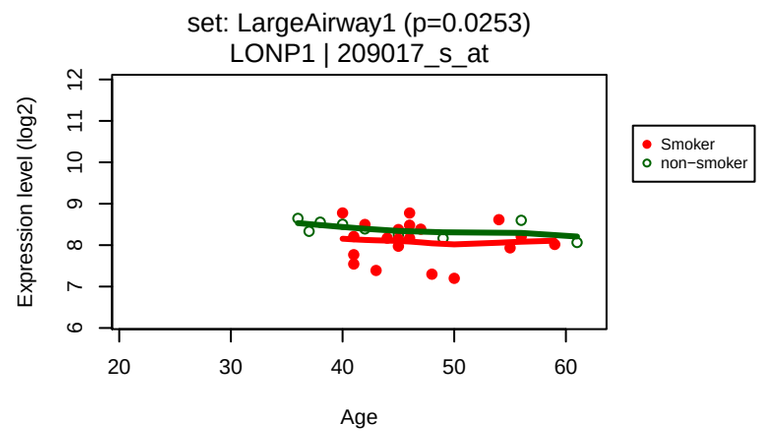

Gene expression of smokers and nonsmokers
gene: LONP1 | 209017_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0215
p=0.5816
p=0.1829
p=0.0253
p=0.4384
p=0.5303
p=0.0113
p=0.1246
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

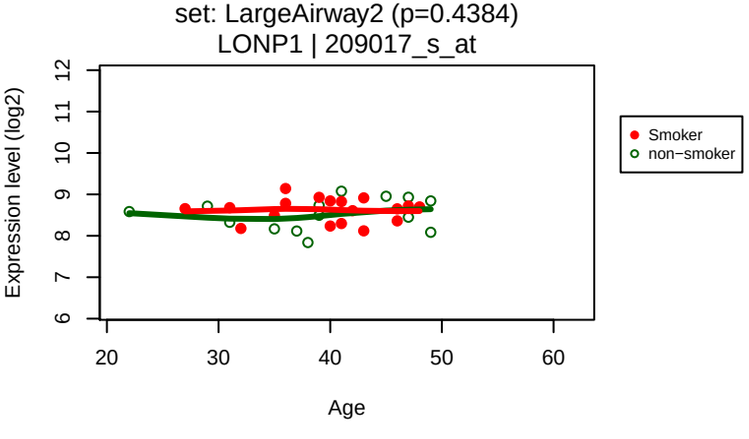

Gene expression ratio (smokers/nonsmokers)
gene: LONP1 | 209017_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0215
p=0.5816
p=0.1829
p=0.0253
p=0.4384
p=0.5303
p=0.0113
p=0.1246
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

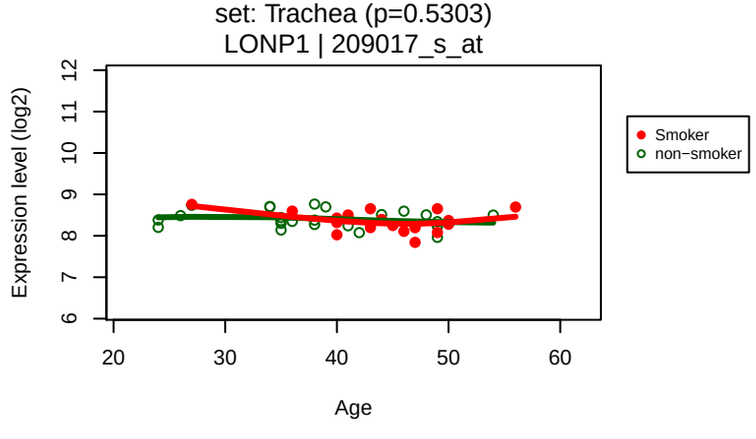

Gene expression ratio in Males
gene: LONP1 | 209017_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2874
p=0.0125
p=0.8356
p=0.0695
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LONP1 | 209017_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4466
p=0.024
p=0.5833
p=0.9453
SmallAirway2
LargeAirway2
Trachea
Macrophage2

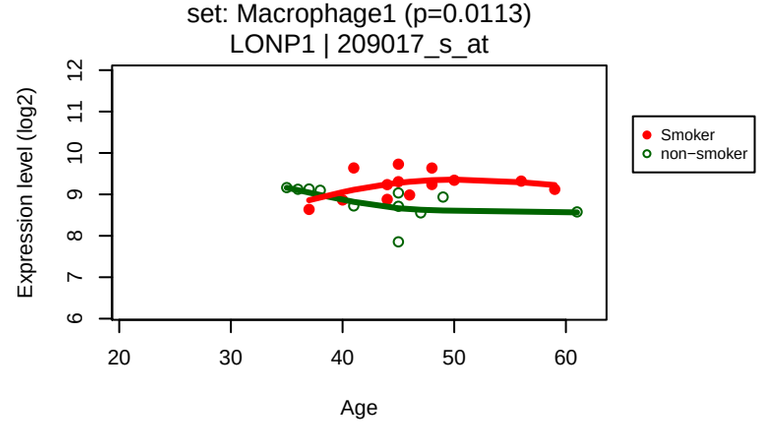

set: SmallAirway1 (p=0.0215)
LONP1 | 209017_s_at
Expression level (log2)
Age
Smoker
non−smoker

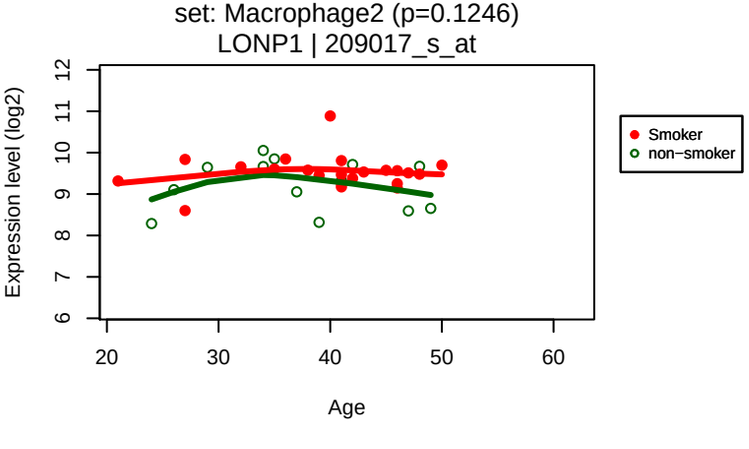

set: SmallAirway2 (p=0.5816)
LONP1 | 209017_s_at
Expression level (log2)
Age
Smoker
non−smoker

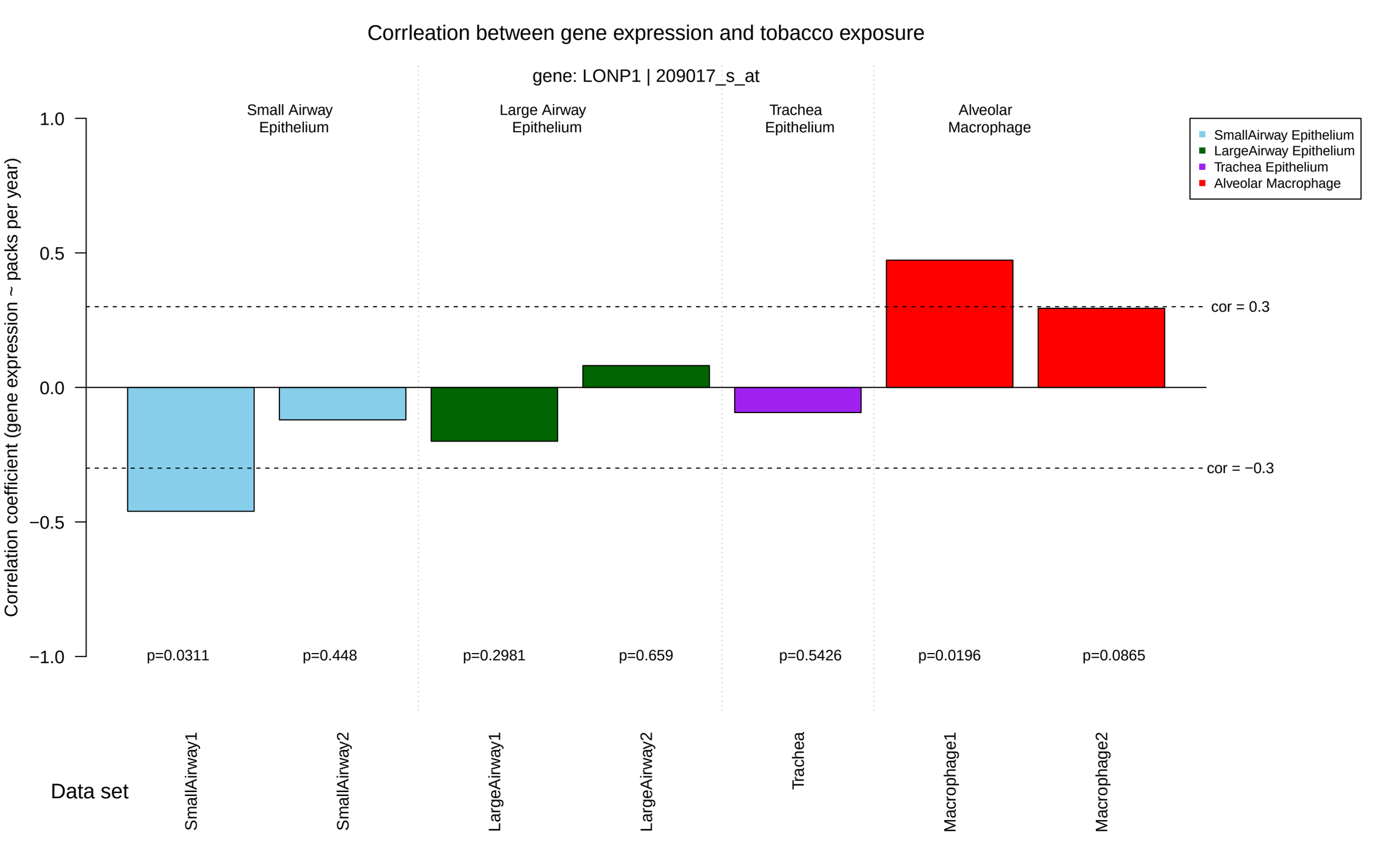

set: LargeAirway1 (p=0.0253)
LONP1 | 209017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4384)
LONP1 | 209017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5303)
LONP1 | 209017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0113)
LONP1 | 209017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1246)
LONP1 | 209017_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LONP1 | 209017_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0311
p=0.448
p=0.2981
p=0.659
p=0.5426
p=0.0196
p=0.0865
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2